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RESEARCH METHODOLOGY

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UNIT I : INTRODUCTION

TAKEAWAYS

- MEANING AND IMPORTANCE
- CLASSIFICATION OF RESEARCH
 - RESEARCH PROCESS
 - RESEARCH FUNCTIONS

THE CONCEPT OF RESEARCH

- Research is an activity that leads us to finding new facts, information, assisting us in verifying the available knowledge and in making us question things that are difficult to understand as per existing data.
- To be successful manager it is important for you to know how to go about making the right decisions by being knowledgeable about the various steps involved in finding solutions to problematic issues.

IT MAY BE UNDERSTOOD IN FOLLOWING TERMS ALSO:

- Research is a continuous activity in majority of disciplines and professions.
- It is helpful in critical assessment of the way we work, execute policies, and give instructions in our professions.
- It is systematic observation of processes to find better ways to do things and to reduce the effort being put in to achieve an objective and identifying the validity of the targets.
- In fact research is a subconscious activity that we are involved in, at all times whether it is purchase of daily use articles , a car, an electronic good or planning a holiday.

MEANING OF RESEARCH

- Research is a process to discover new knowledge to find answers to a question. The word research has two parts re (again) and search (find) which denote that we are taking up an activity to look into an aspect once again or we want to look for some new information about something.
- E.g Front Office Executive has to learn about the facilities, timings, key features of products and services available at the hotel if one wants to become a wonderful sales professional.
- "All progress is born of inquiry. Doubt is often better than overconfidence, for it leads to inquiry, and inquiry leads to invention" in context of which the significance of research can well be understood.

- According to Thyer (2001), "the word research is composed of two syllables, re and search. re is a prefix meaning again, anew or over again, search is a verb meaning to examine closely and carefully, to test and try, or to probe. Together they form a noun describing a careful, systematic, patient study and investigation in some field of knowledge, undertaken to establish facts or principles."

- According to Rocco (2011), "Research is a careful investigation or inquiry especially through search for new facts in any branch of knowledge."
- Redman and Mory (2010), "Systematized effort to gain new knowledge."
- Kothari, C.R, "Research is manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art."
- Creswell (2008), "Research is systematic investigation to establish the facts."

RESEARCH AS A SCIENTIFIC PROCESS

- Research is simply the process of finding solutions to a problem after a targeted and systematic study and analysis of materials and sources.
- It is a prearranged / structured enquiry (a formal step by step method or sequence to take up research activity is developed to ensure correctness of data and validity of processes).
- Scientific methods consist of systematic observation, classification and interpretation of data. It utilizes acceptable scientific methodology to solve problems (the method used should be able to give repetitive results under similar conditions)

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- It should create new knowledge that is generally applicable. (The outcomes should be such that they are not specific to particular issue or a situation but need to be generalized for application to comparable issues).
 - It is creative process to develop better understanding of mankind, social and cultural and economical issues.
 - It should be useful to others who wish to apply the findings in developing new policies or applications of findings of a research in the benefit of public.

OBJECTIVES OF RESEARCH

- Kothari (1990): To discover answers to questions through the application of scientific methods. To generate new questions and new solutions with new times.
- Other relevant objectives:
 - To discover new facts
 - To verify and test old facts
 - To develop new Scientific tools concepts and theories
 - To know connections between human activities
 - To understand the function of society
 - To explore social reality

THE HALLMARKS OF SCIENTIFIC RESEARCH

- The main distinguishing characteristics of scientific research may be listed as:
 - Purposiveness
 - Research should have a purpose focus
 - Rigor
 - Rigor research involves a good theoretical base and a careful thought out methodologies
 - Testability
 - Research tends to test logically developed hypothesis
 - Replicability
 - When the result from one study is found to be consistent by another study, it is more likely to represent a reliable claim to new knowledge

- Precision and confidence
 - Precision refers to the closeness of the findings to the reality based on a sample. While confidence refers to the probability that our estimations are correct.
- Objectivity
 - Research should be based on facts derived from actual data.
- Generalizability
 - Scope of applicability of the research findings in different settings
- Parsimony
 - Economy in research models is achieved when we can build into our research framework a lesser number of variables that explain the variance far more efficiently than the complex set of variables.

TOP COMPANIES INVESTMENTS IN R&D

- **Amazon (AMZN), \$42.74 billion**

- Amazon is among the top R&D spenders. Amazon reveals a whopping expenditure of \$42.74 billion in the fiscal 2020 (11.1% of net sales) on ‘technology and content’ as compared to \$35.93 billion in the fiscal 2019.

- **Alphabet (GOOG, GOOGL), \$27.57 billion**

- Alphabet spent \$27.57 billion on R&D, which is equivalent to 15.1% of its revenue of \$182.57 billion during the fiscal 2020. The company’s R&D spending has more than doubled since the fiscal 2016.

- **Huawei, \$22.04 billion**

- Huawei is a leading global provider of information and communications technology (ICT) infrastructure and smart devices. During the fiscal 2020, it spent around ¥141.893 billion on R&D, which is equivalent to \$22.04 billion. Its R&D spending constituted 15.9% of its total revenue of ¥891.37 billion in fiscal 2020.

- **Microsoft (MSFT), \$27. 2 billion**

- The R & D investment as of December 31, 2023 were \$27.2, a 3.37% increase year-over-year. Microsoft annual research and development expenses for 2023 were \$27.524, a 10.95% increase from 2022

- **Apple (AAPL), \$18.75 billion**

- During fiscal 2020 (Apple's fiscal year runs from October 1 to September 30), Apple spent \$18.75 billion on R&D, equivalent to 7% of its net sales. In the first six months of fiscal 2021 (October 2020 – March 2021), Apple reported its R&D spending at \$10.42 billion.

WHAT IS A RESEARCH ENGINEER?

- Research engineers typically work within the Research & Development department of an organization, government agency or academic institution. Generally, research engineers develop products, processes or technology for their employer.
- To achieve this, they gather relevant information, data or samples, then analyze their research and perform tests to create optimal and innovative solutions.

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• FIELDS WHERE RESEARCH ENGINEERS WORK INCLUDE:

- Medical or health care
- Transportation
- Military
- Computer hardware and software
- Product development (industrial and commercial)
- Energy (oil & gas, renewable energy, mining, etc.)

WHAT DOES A RESEARCH ENGINEER DO?

- Conducting research to identify solutions for industry problems
- Develop concepts for existing or new products, processes or equipment relevant to their industry
- Design industry-relevant products, equipment or technology based on concepts and ensuring functionality
- Building product and system prototypes for testing purposes
- Using specialized equipment to analyze and test processes, equipment or materials
- Using statistical processes to assess data
- Manage and lead members of design or research teams
- Coordinate timelines, resources and tasks to achieve project goals as a project leader
- Preparing reports summarizing tests and their findings
- Writing research and grant proposals

RESEARCH PROCESS

- All research endeavors share a common goal of furthering our understanding of the problem and thus all traverse through certain basic stages, forming a process called the research process.
- An understanding of the research process is necessary to effectively carry out research and sequencing of the stages inherent in the process.

- The Research Process is a process of multiple scientific steps in conducting the research work. Each step is interlinked with other steps. The process starts with the research problem at first. Then it advances in the next steps sequentially. Generally, a researcher conducts research work within eight steps
- Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps. The chart shown in Figure well illustrates a research process. The chart indicates that the research process consists of a number of closely related activities.



RESEARCH PROBLEM

- Any question that you want to answer or any assumption or statement that you want to challenge or investigate or examine can become a research problem.
- Questions to answer
 - Is there any relationship between technology and business?
 - What are the effects of TV advertisement on customers behavior?

THE IMPORTANCE OF FORMULATING A RESEARCH PROBLEM

- Formulation of a research problem is the first and most important step of the research process.
- It is like the identification of a destination before undertaking a journey.
- In the absence of a clear research problem, a clear and economical plan is impossible.
- Therefore; a research problem is like the foundation of a building.
- The type and design of the building is dependent upon the foundation.
- If the foundation is well designed and strong you can expect the building to be also.
- According to Kerlinger :
 - If one wants to solve a problem, one must generally know what the problem is. It can be said that a large part of the problem lies in knowing what one is trying to do (1986:17).

CONSIDERATIONS IN SELECTING A RESEARCH PROBLEM

- Interest
 - You must select a topic that interests you most. If you select a topic which does not greatly interest you, it could become extremely difficult to sustain the required motivation, and hence its completion as well as the amount of time taken could be affected.
- Magnitude
 - You should have sufficient knowledge about the research process to be able to visualize the work involved in completing the proposed topic.
- Measurement of concepts
 - If you are using a concept in your study make sure you are clear about its indicators and their measurements. Do not use concepts in your research problem that you are not sure how to measure.
- Level of expertise
 - Make sure you have an adequate level of expertise for the task you are proposing. Allow for the fact that you will learn during the study and may receive help from your research supervisors and others, but remember you need to do most of the work yourself.

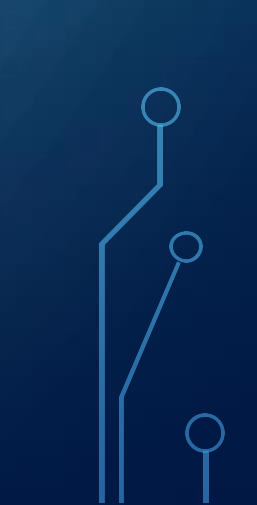
- **Relevance**

- Select a topic that is relevant to you as a professional. Ensure that your study adds to the existing body of knowledge, bridges current gaps or is useful in policy formulation. This will help you to sustain interest in the study.

- **Availability of data**

- If your topic requires collection of information from secondary sources (office records, client records, census or other already- published reports, etc.) before finalizing your topic, make sure that these data are available and in the format you want.

- **Ethical issues**

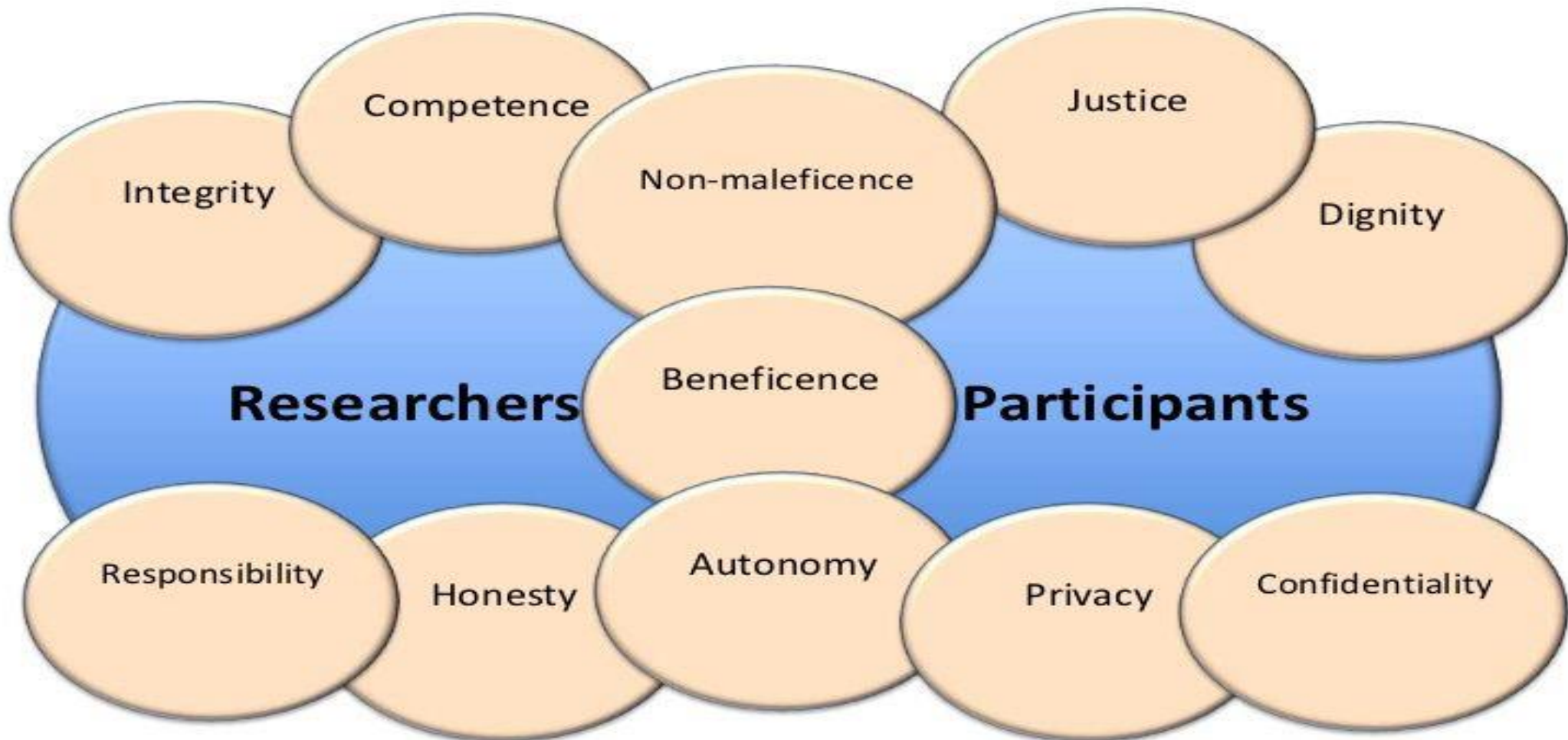
- In a situation where your topic deals with sensitive issues, it is important to determine how ethical issues can affect the study population, and how ethical problems can be over-come should be thoroughly examined at the problem-formulation stage.
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ISSUES GOVERNING RESEARCH FUNCTION

- Ethics are a personal code of conduct based on respect for one's self, others and surroundings and is governed by the principles or assumptions underpinning the way individuals or organisations ought to conduct themselves.
- Research ethics involves the application of fundamental ethical principles to research activities which include the design and implementation of research, respect towards society and others, the use of resources and research outputs, scientific misconduct and the regulation of research.

PRINCIPLES OF RESEARCH ETHICS

Ethical Principles of Research



THE IMPORTANCE OF RESEARCH ETHICS

- They promote the aims of research, such as expanding knowledge.
- They support the values required for collaborative work, such as mutual respect and fairness. This is essential because scientific research depends on collaboration between researchers and groups.
- They mean that researchers can be held accountable for their actions. Many researchers are supported by public money, and regulations on conflicts of interest, misconduct, and research involving humans or animals are necessary to ensure that money is spent appropriately.
- They ensure that the public can trust research. For people to support and fund research, they have to be confident in it.
- They support important social and moral values, such as the principle of doing no harm to others.
 - **Source:** Resnick, D. B. (2015) *What is Ethics in Research and Why is it Important?*

CLASSIFICATION OF RESEARCH

- Research can be classified in terms of
 - Goal of research
 - Basic Research
 - Applied research
 - Approaches of research
 - Qualitative research
 - Quantitative research
 - The type of data used
 - Primary research
 - Secondary research

BASIC RESEARCH

- Basic (aka fundamental or pure) research is driven by a scientist's curiosity or interest in a scientific question. The main motivation is to expand man's knowledge, not to create or invent something. There is no obvious commercial value to the discoveries that result from basic research.
- For example, basic science investigations probe for answers to questions such as:
 - How did the universe begin?
 - What are protons, neutrons, and electrons composed of?
 - What is the specific genetic code of the fruit fly?

APPLIED RESEARCH

- Applied research refers to scientific study and research that seeks to solve practical problems. Applied research is used to find solutions to everyday problems, cure illness, and develop innovative technologies, rather than to acquire knowledge for knowledge's sake.
- For example, applied researchers may investigate ways to:
 - Improve agricultural crop production
 - Treat or cure a specific disease
 - Improve the energy efficiency of homes, offices, or modes of transportation

CORRELATIONAL RESEARCH

- Correlational research refers to the systematic investigation or statistical study of relationships among two or more variables, without necessarily determining cause and effect.
- It Seeks to establish a relation/association/correlation between two or more variables that do not readily lend themselves to experimental manipulation.
- For example, to test the hypothesis “ Listening to music lowers blood pressure levels” there are 2 ways of conducting research
- Experimental – group samples and make one group listen to music and then compare the bp levels
- Survey – ask people how they feel ? How often they listen? And then compare

DESCRIPTIVE RESEARCH

- Descriptive research refers to research that provides an accurate portrayal of characteristics of a particular individual, situation, or group. Descriptive research, also known as statistical research.
- These studies are a means of discovering new meaning, describing what exists, determining the frequency with which something occurs, and categorizing information.
- In short, descriptive research deals with everything that can be counted and studied, which has an impact of the lives of the people it deals with.
- For example,
 - finding the most frequent disease that affects the children of a town. The reader of the research will know what to do to prevent that disease thus, more people will live a healthy life.

EXPERIMENTAL RESEARCH

- Experimental research is an objective, systematic, controlled investigation for the purpose of predicting and controlling phenomena and examining probability and causality among selected variables.
- The experimental research design must be carried out in a controlled environment. Throughout the experiment, the researcher collects data that can support or refute a hypothesis, thus, this research is also referred to as hypothesis testing or a deductive research method.

EXPLORATORY RESEARCH

- Exploratory research is a type of research conducted for a problem that has not been clearly defined. Exploratory research helps determine the best research design, data collection method and selection of subjects.
- The results of exploratory research are not usually useful for decision-making by themselves, but they can provide significant insight into a given situation
- Exploratory research is not typically generalizable to the population at large.
- Exploratory research can be quite informal, relying on secondary research such as reviewing available literature and/or data, or qualitative approaches such as informal discussions with consumers, employees, management or competitors, and more formal approaches through in-depth interviews, focus groups, projective methods, case studies or pilot studies.

HISTORICAL RESEARCH

- Historical research is research involving analysis of events that occurred in the remote or recent past
- Application
- Historical research can show patterns that occurred in the past and over time which can help us to see where we came from and what kinds of solutions we have used in the past.
- Understanding this can add perspective on how we examine current events and educational practices.
- The steps involved in the conduct of historical research
- Here are the five steps:
 1. Identification of the research topic and formulation of the research problem or question.
 2. Data collection or literature review
 3. Evaluation of materials
 4. Data synthesis
 5. Report preparation or preparation of the narrative exposition

QUALITATIVE RESEARCH

- Qualitative research is research dealing with phenomena that are difficult or impossible to quantify mathematically, such as beliefs, meanings, attributes, and symbols
- Qualitative researchers aim to gather an in-depth understanding of human behaviour and the reasons that govern such behaviour. The qualitative method investigates the why and how of decision making, not just what, where, when.

ADVANTAGES & DISADVANTAGES

- Advantages

- It enables more complex aspects of a person's experience to be studied
- Fewer restrictions or assumptions are placed on the data to be collected.
- Not everything can be quantified, or quantified easily, Individuals can be studied in more depth
- Good for exploratory research and hypothesis generation
- The participants are able to provide data in their own words and in their own way

- Disadvantages

- It is more difficult to determine the validity and reliability of linguistic data
- there is more subjectivity involved in analysing the data. • “Data overload” – open-ended questions can sometimes create lots of data, which can take a long time to analyse!
- Time consuming

QUANTITATIVE RESEARCH

- Quantitative research refers to the systematic empirical investigation of any phenomena via statistical, mathematical or computational techniques. The objective of quantitative research is to develop and employ mathematical models, theories and/or hypotheses pertaining to phenomena
- Quantitative research is generally made using scientific methods, which can include:
 - The generation of models, theories and hypotheses
 - The development of instruments and methods for measurement
 - Experimental control and manipulation of variables
 - Collection of empirical data
 - Modelling and analysis of data
 - Evaluation of results

ADVANTAGES & DISADVANTAGES

- Advantages

- Quantitative research allows the researcher to measure and analyse data.
- The researcher is more objective about the findings of the research.
- Quantitative research can be used to test hypotheses in experiments because of its ability to measure data using statistics.

- Disadvantages

- The main disadvantage of quantitative research is the context of the study or experiment is ignored.
- Quantitative research does not study things in a natural setting or discuss the meaning things have for different people.
- A large sample of the population must be studied for more accurate results

TYPE OF DATA

- Depending on the type of data generated and used research can be classified as
 - Primary (field research)
 - Secondary research (desk research).
- Primary research involves generating primary data or the collection of data that does not already exist whereas secondary research involves the summery and synthesis of existing data

ADVANTAGES OF PRIMARY DATA

- Original and independent collection increased the authenticity of data.
- Directly Collection enhance the reliability of data.
- Used in both quantitative and qualitative research methods.
- Hidden information can be collected through primary data.
- After analyzed primary data can be used as secondary data.

DISADVANTAGES OF PRIMARY DATA

- Reliability depend on respondents information accuracy.
- Information can be bias.
- Expensive and time consuming in nature.
- Lack of experience among researcher.
- Requires field work.

ADVANTAGES OF SECONDARY DATA

- These data can be quickly manageable.
- Time and cost balance remains maintained.
- Information available is already analyzed by experts.
- Used to update data or reinterpret existing ones.
- Helpful for philosopher, thinker or authors for developing new concept.
- No or less Field work.

DISADVANTAGES OF SECONDARY DATA

- No standard measurement of validity.
- Need expertise.
- Accuracy and reliability is always lesser than primary data.